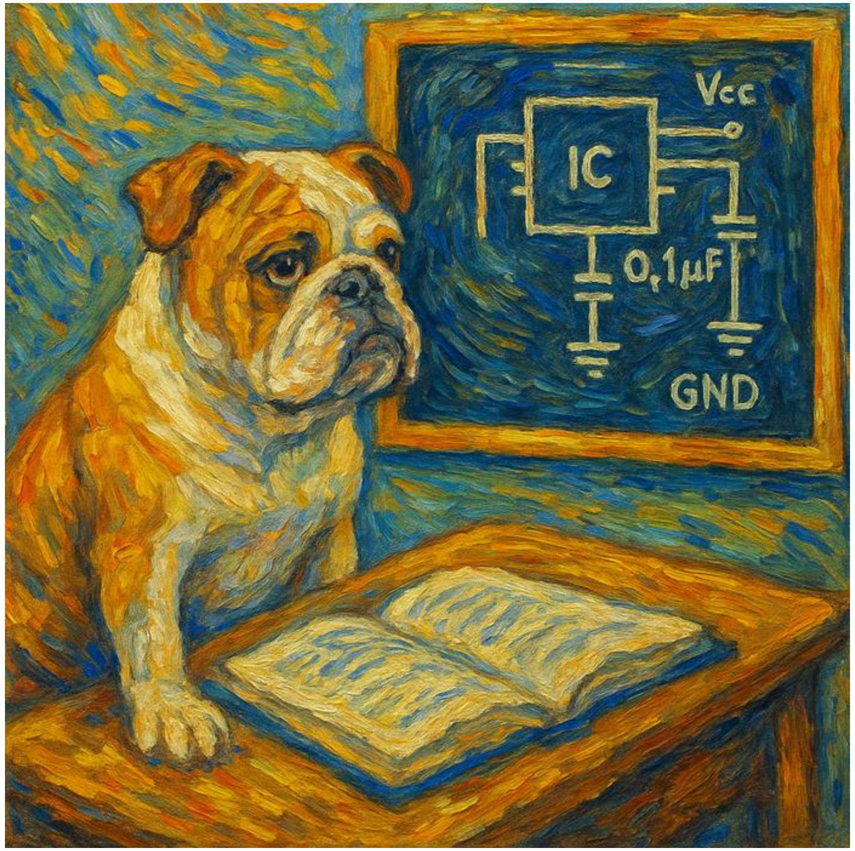
Hands-on PCB Design and Electronic Prototyping with Autodesk Fusion

SUMMER ENRICHMENT- LYLES COLLEGE OF ENGINEERING REVISION 1.0

Day 6: Review

- Advanced PCB Design and Best Practices. These techniques would also be applicable to IC design.
  - Signal Integrity and EMC
    - Managing crosstalk and noise
    - Proper grounding and decoupling capacitors
  - High-speed and RF considerations
  - Hands-on Activity
    - Use AI to recommend possible simple PCB projects
    - Select one and start designing



Day 7: PCB Design Verification and Manufacturing Prep

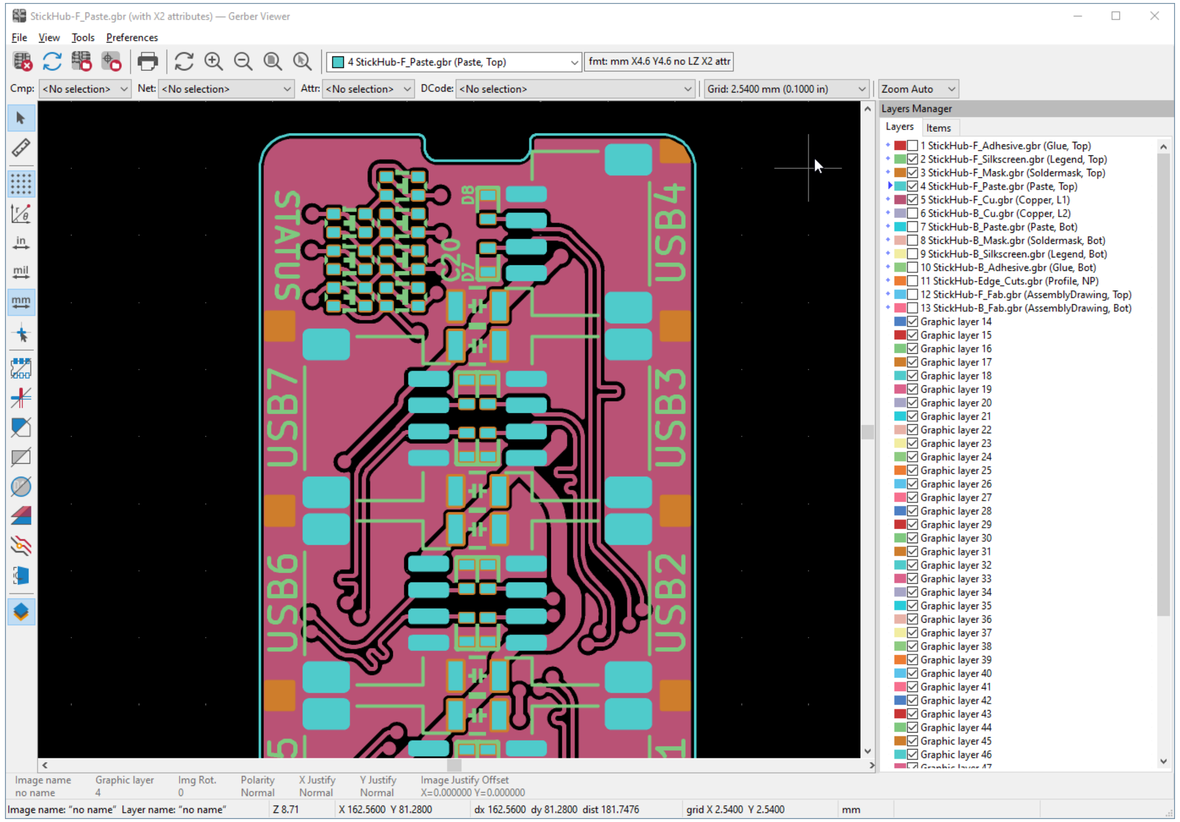
- Design Rule Checks (DRC)
  - Setting up rules and constraints
  - Identifying and correcting common errors
- Generating Fabrication Files
  - Gerber file generation and drill files
  - Bill of Materials (BOM) and assembly documentation
- Hands-on Activity
  - Generating manufacturing outputs for previous design
  - Reviewing Gerber and Drill files using KiCad Gerber Viewer
  - Continued work on PCB Project
  - Soldering Overview Tomorrow
  - We have some PCB’s finished!!!

AI Minute

- Please type the following prompts into the AI Tool of your choice, Chat GPT, Gemini, Grok, Copilot, etc…
  - **Can you compare the Raspberry Pi Pico to the Arduino Nano? Which one is better for University Engineering Students, and which is better for High School Students?**
  - **For the previous prompt, create a decision matrix chart.**

KiCad Gerber Viewer

- KiCad has a Gerber Viewer
  - View
    - Gerber Files
    - Drill Files (Excellon)
- Download and Install KiCad
  - https://www.kicad.org/download/

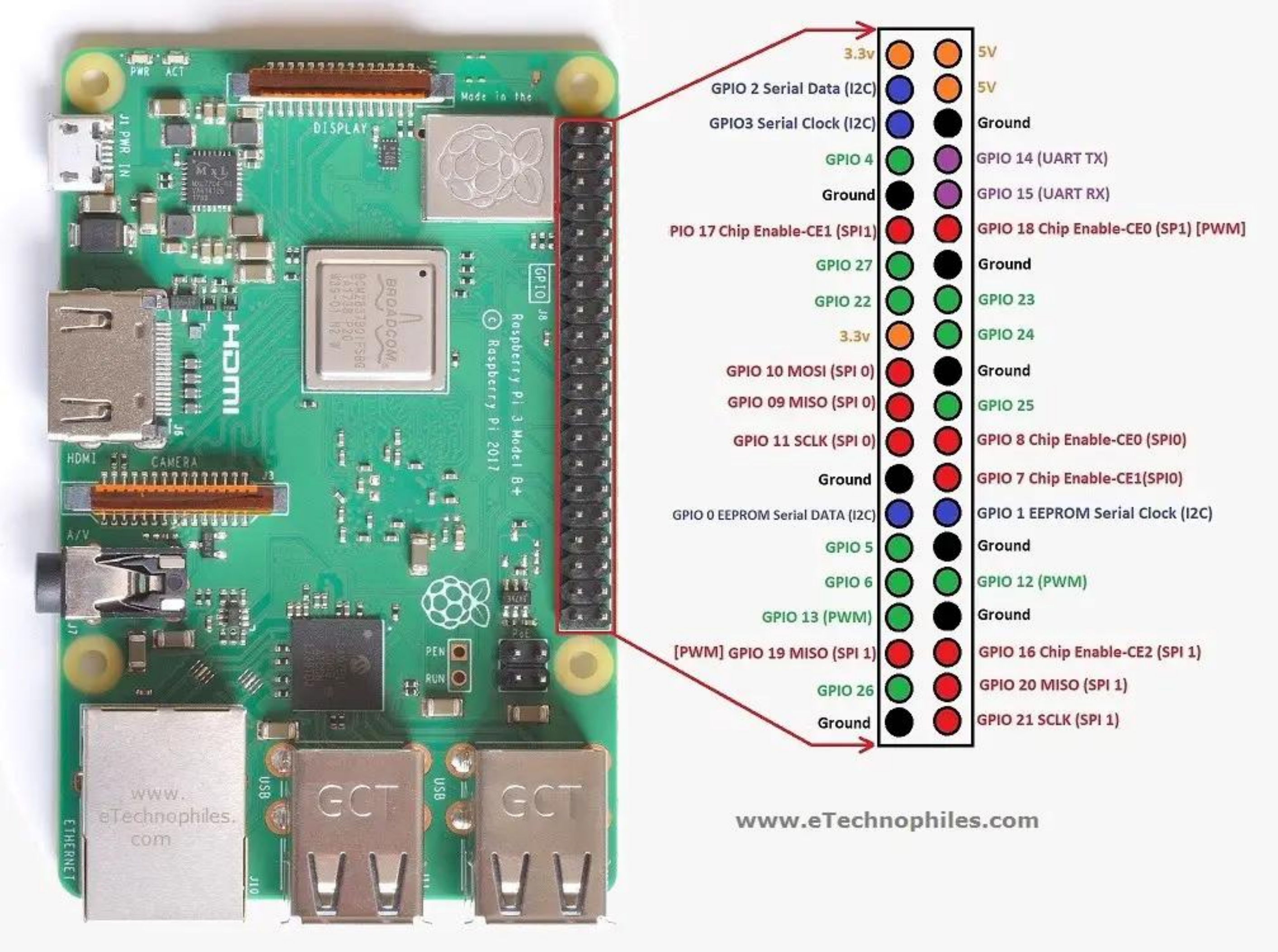


AI Minute

- Please type the following prompts into the AI Tool of your choice, Chat GPT, Gemini, Grok, Copilot, etc…
  - **Please create a list of 100 project ideas for beginning university-level electrical engineering students to design a very simple, but useful, PCB?**
  - **Select a project on the list, or choose another project, and type something to the effect of- "Can you show a circuit for (selected project), list what parts are needed and describe how it works?"**

Raspberry Pi Pinout- Create Library Component- PCB Footprint

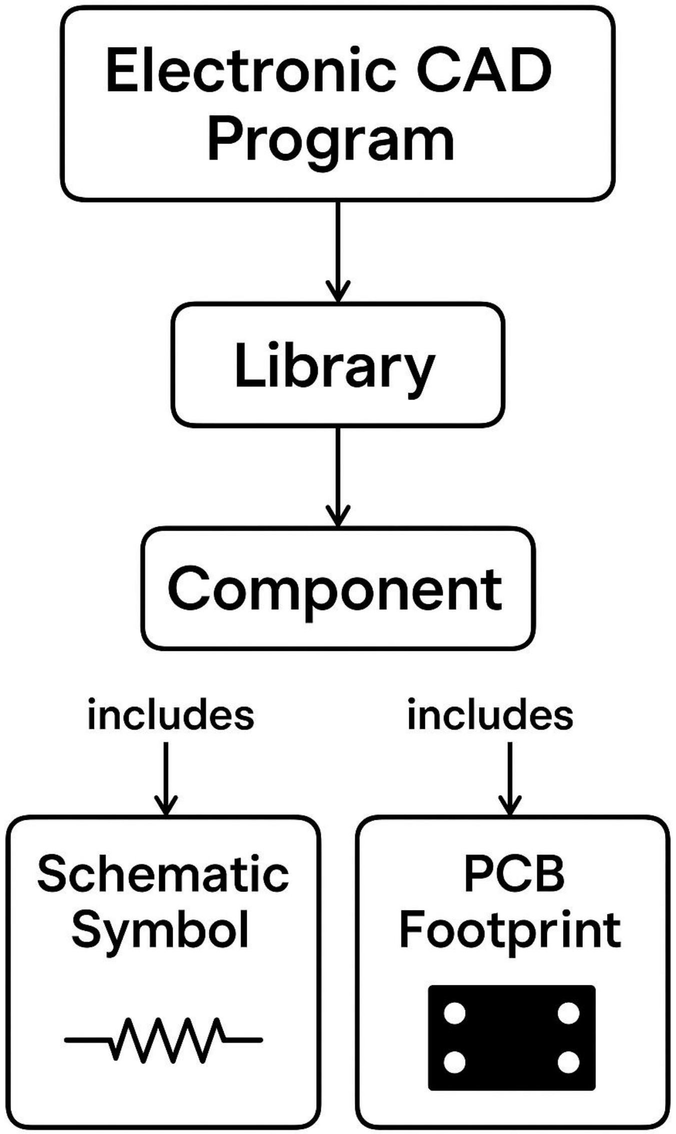
- Component Creation
  - Create Schematic Symbol
  - Create PCB Footprint
  - Combine Footprint and Symbol into a Component
- Create a new project
- Place the newly created Raspberry Pi I/O connector component on the schematic and run ERC
- Switch to PCB and place the component on the PCB
- Adjust the PCB outline to be a more appropriate size for the connector




3.3v
5V
GPIO 2 Serial Data (I2C)
5V
GPIO3 Serial Clock (I2C)
Ground
GPIO 4
GPIO 14 (UART TX)
Ground
GPIO 15 (UART RX)
PIO 17 Chip Enable-CE1 (SPI1)
GPIO 18 Chip Enable-CE0 (SP1) [PWM]
GPIO 27
Ground
GPIO 22
GPIO 23
3.3v
GPIO 24
GPIO 10 MOSI (SPI 0)
Ground
GPIO 09 MISO (SPI 0)
GPIO 25
GPIO 11 SCLK (SPI 0)
GPIO 8 Chip Enable-CE0 (SPI0)
Ground
GPIO 7 Chip Enable-CE1(SPI0)
GPIO 0 EEPROM Serial DATA (I2C)
GPIO 1 EEPROM Serial Clock (I2C)
GPIO 5
Ground
GPIO 6
GPIO 12 (PWM)
GPIO 13 (PWM)
Ground
[PWM] GPIO 19 MISO (SPI 1)
GPIO 16 Chip Enable-CE2 (SPI 1)
GPIO 26
GPIO 20 MISO (SPI 1)
Ground
GPIO 21 SCLK (SPI 1)

CAD Library Elements

- Libraries
  - Libraries contain components
- Components
  - Components include:
    - A Schematic Symbol
    - A PCB Footprint
    - A 3D Model



Questions?